

LITERATURE SYNTHESIS

Accuracy, Calibration, Autonomy, and Physical Integration of Low-Cost Outdoor Sensor Boxes

Summary of the 19 research sources currently included in the Enclosure-Research repository

Prepared for	Principal Investigator review
Working paper	Calibration, Ambient Conditions, and Integration Effects on the Accuracy and Autonomy of a Low-Cost Outdoor Multi-Sensor Box
Date	June 11, 2026

Purpose

This report connects each literature source to the proposed study, summarizes the reported sensors, materials, and geometries, and identifies evidence-supported considerations for the next experimental phase.

Executive Summary

The literature supports framing the lab's sensor box as an integrated measurement instrument rather than as a collection of individual sensors or an enclosure-only design. Field accuracy depends on calibration, but it also depends on airflow, direct and reflected radiation, sensor placement, electronics self-heating, moisture protection, power, communication, storage, and maintenance.

The strongest recurring result is that passive shields fail most noticeably during high-solar, low-wind conditions. Stacked plates, overlapping louvers, reflective surfaces, and separated electronics improve performance without consuming power. Active aspiration can further reduce airflow-dependent bias, but the fan directly reduces autonomy and adds a failure mode. Long-duration deployments consistently show that connectors, mounts, batteries, charging, storage, communications, and firmware may fail before the sensing element itself.

Implications for the Proposed Paper

- Begin with the current lab box as an unchanged baseline and compare it with reference instruments.
- Separate the sealed electronics/power region from sensors that require ambient airflow.
- Evaluate raw and calibrated accuracy, but do not use calibration to conceal avoidable enclosure heating or airflow problems.
- Report autonomy using runtime before intervention, expected versus received samples, uptime, maintenance events, and failure types.
- Use a small, justified design comparison: current box, strong passive shield, and optional aspirated benchmark.
- Treat material, color/finish, geometry, airflow, sealing, and sensor-to-electronics distance as measurable factors.

Cross-Paper Findings

Sensor selection

- Digital temperature/RH sensors are compact and easy to integrate, but their readings are strongly affected by shielding, airflow, condensation, and placement.
- Optical PM and electrochemical gas sensors require environmental correction, repeated calibration, and controlled exposure to ambient air.
- Reference-grade Pt100 studies provide strong shield-comparison methods, even when the final lab sensor is lower cost.

Material and geometry

- White/off-white ASA is the best-supported printed outdoor material; PLA is useful for prototypes but requires durability caution.
- Reflective foil and white corrugated plastic provide very low-cost geometry benchmarks.
- Passive stacked plates generally outperform open or weakly ventilated tube geometries.
- IP-rated boxes are appropriate for electronics, but ambient sensors should be external or in a separately ventilated region.

Autonomy and reliability

- Battery life alone is not autonomy; physical intervention can be triggered by water, corrosion, memory, fan, firmware, RF, connector, or mounting failures.
- Pre/post co-location and device-health logging improve the ability to identify drift and limiting failures.

Literature at a Glance

The table below identifies the primary contribution of each source. Detailed source profiles follow.

#	Paper	Primary contribution	Most relevant design issue
1	Theisen et al. (2020)	Complete system / autonomy	The sensing element is only one part of field accuracy and lifetime. Integration and maintenance failures can dominate.
2	Tatsumi et al. (2021)	Complete system / autonomy	Low-cost, modular hardware is practical, but exact material reporting and external-connection protection are necessary for reproducibility.
3	Grimsley et al. (2022)	Complete system / autonomy	A modular sealed logger can support long deployments when sensing, power, communications, and maintenance are designed together.
4	Daepf et al. (2022)	Complete system / autonomy	Completeness and remote serviceability are central measures of whether a low-cost box is useful in practice.
5	Lazarescu (2015)	Complete system / autonomy	Theoretical battery life does not establish autonomy; observed mechanical, thermal, RF, and firmware behavior does.
6	Botero-Valencia et al. (2022)	Shield geometry / material	A modular passive cone-stack shield is a defensible low-cost baseline, but material and color were not fully isolated.
7	Tarara and Hoheisel (2007)	Shield geometry / material	Radiation blocking alone is insufficient: reflected radiation and ventilation area determine passive-shield error.
8	Holden et al. (2013)	Shield geometry / material	Simple, inexpensive geometry can be effective enough for distributed sensing, but passive limitations remain.
9	Liu et al. (2023)	Shield geometry / material	Mixed surface finishes and guided natural ventilation offer performance gains, with added fabrication and coating-maintenance complexity.
10	García Izquierdo et al. (2024)	Shield geometry / material	Long-duration comparison must evaluate operational reliability as well as thermometer error.
11	Deford et al. (2025)	Shield geometry / material	Active airflow is a useful accuracy benchmark and a direct test of the accuracy-autonomy tradeoff.
12	Jin et al. (2026)	Shield geometry / material	Advanced passive geometry may reduce error without fan power, but complexity and weathering must be assessed.
13	deSouza et al. (2022)	Calibration / drift	Calibration performance must be

#	Paper	Primary contribution	Most relevant design issue
			tested beyond the location and period used to fit the model.
14	Clements et al. (2017)	Calibration / drift	Low-cost sensor boxes need a defined quality-assurance workflow rather than a one-time validation.
15	Giordano et al. (2021)	Calibration / drift	PM calibration and enclosure design are deployment-specific and cannot be treated independently.
16	Vajs et al. (2021)	Calibration / drift	Interpretable linear and multiple-linear models provide a defensible baseline before more complex methods.
17	Schneider et al. (2023)	Calibration / drift	Pre- and post-deployment co-location improves detection of drift and field-induced change.
18	Diez et al. (2024)	Calibration / drift	Long-term validation requires documentation of hardware, firmware, processing, location, and uncertainty.
19	Winter et al. (2025)	Calibration / drift	Recalibration interval and physical-maintenance interval are separate engineering quantities.

Complete Sensor-Box and Field-System Studies

These papers show how sensors, packaging, power, communication, and maintenance behave as one deployed system.

1. Theisen et al. (2020)

More Science with Less: Evaluation of a 3D-Printed Weather Station

DOI: <https://doi.org/10.5194/amt-13-4699-2020>

Sensors	MCP9808; HTU21D; BMP280; SI1145; Hall-effect wind, direction, and rain sensors.
Materials	Off-white/gray ASA; PVC frame; electrical junction box; polyurethane, PTFE filter, conformal coating, and silicone.
Geometry	Integrated 3D-PAWS station; naturally aspirated shield; crossbar-mounted wind/UV sensors; separate junction box for logger.
Experimental setup	Eight-month outdoor deployment beside an Oklahoma Mesonet reference station.

What it shows: Several low-cost measurements can be useful, but wind dependence, corrosion, moisture, connectors, printed fasteners, and moving parts strongly affect total system performance.

Brief conclusion: The sensing element is only one part of field accuracy and lifetime. Integration and maintenance failures can dominate.

Importance to this research: Provides the closest precedent for evaluating the lab box as a complete deployed instrument and for recording both accuracy and failure events.

Things to consider moving forward

- Log internal box temperature and wind speed so radiation-shield performance can be separated from sensor calibration.
- Use metal structural fasteners and explicitly inspect connectors, drain paths, coatings, and exposed printed parts.

2. Tatsumi et al. (2021)

An Open-Source, Low-Cost Measurement System for Collecting Hydrometeorological Data in the Open Field

DOI: <https://doi.org/10.3390/technologies9040078>

Sensors	LM60BIZ air/soil temperature; YL-69 soil moisture; G2711-01 GaAsP photodiode; GPS and micro-SD.
Materials	3D-printed protective case using unspecified resin; handmade photodiode cover.
Geometry	Compact rectangular logger case with external sensors connected through detachable terminals; solar-powered system.
Experimental setup	Grass-field deployment with reference comparisons; 30-day temperature and 19-day soil-moisture/PPFD measurements.

What it shows: An open-field system must be evaluated as a package of sensing, power, storage, enclosure, and external connections.

Brief conclusion: Low-cost, modular hardware is practical, but exact material reporting and external-connection protection are necessary for reproducibility.

Importance to this research: Supports the baseline inventory and reinforces documenting the complete lab box before modification.

Things to consider moving forward

- Record exact material chemistry rather than only stating that a case is 3D printed.
- Include connector sealing, strain relief, and external-probe replacement in maintainability testing.

3. Grimsley et al. (2022)

Experiences in LP-IoT: EnviSense Deployment of Remotely Reprogrammable Environmental Sensors

DOI: <https://doi.org/10.1145/3477085.3478988>

Sensors	MaxBotix ultrasonic range sensor or SDI-12 pressure transducer; internal T/RH/pressure; battery voltage/current; GPS; LoRa.
Materials	Commercial IP67 enclosure; exact enclosure material not reported.
Geometry	Sealed rectangular logger with four sealed ports and external measurement probes.
Experimental setup	More than 12 nodes at two sites; six months of continuous field measurement discussed.

What it shows: Sealed electronics, external probes, remote management, and device-health sensing improve autonomy and diagnosability.

Brief conclusion: A modular sealed logger can support long deployments when sensing, power, communications, and maintenance are designed together.

Importance to this research: Strong precedent for a two-zone architecture and for treating battery and communication health as measurement-system outputs.

Things to consider moving forward

- Separate protected electronics from ambient-air sensors.
- Log battery, resets, and communication health alongside environmental data.

4. Daepf et al. (2022)

Eclipse: An End-to-End Platform for Low-Cost, Hyperlocal Environmental Sensing in Cities

DOI: <https://doi.org/10.1109/IPSNS4338.2022.00010>

Sensors	PM2.5; temperature; RH; pressure; optional O3, NO2, SO2, and CO modules.
Materials	Custom weatherproof enclosure; exact material and airflow path not reported.
Geometry	Modular solar-powered, cellular-connected urban sensor box.
Experimental setup	115 devices deployed across Chicago; more than 90% of expected sensor-hours collected during the reported period.

What it shows: Large-scale deployment is possible when the platform is modular and expected sensor-hours are treated as an operational metric.

Brief conclusion: Completeness and remote serviceability are central measures of whether a low-cost box is useful in practice.

Importance to this research: Provides a clear autonomy metric and supports designing reset/service actions that do not require opening the box.

Things to consider moving forward

- Report expected versus received sensor-hours.
- Evaluate site-specific mounting and solar-panel heating effects.

5. Lazarescu (2015)

Design and Field Test of a WSN Platform Prototype for Long-Term Environmental Monitoring

DOI: <https://doi.org/10.3390/s150409481>

Sensors	NTC temperature transducer; low-power radio; battery and system-health measurements.
Materials	Protective resin; compact node package; sealed plastic gateway package inside a divided wooden housing.
Geometry	Downward-facing vented node; solar-powered gateway with separated compartments.
Experimental setup	Fifty sensor nodes and two gateways deployed for approximately 2.5 months in an olive field.

What it shows: Solar heating, solder fatigue, antenna placement, charger failure, RF attenuation, and faulty-node traffic can limit autonomy.

Brief conclusion: Theoretical battery life does not establish autonomy; observed mechanical, thermal, RF, and firmware behavior does.

Importance to this research: Directly informs the paper's failure-mode and maintenance-event framework.

Things to consider moving forward

- Test antenna clearance and enclosure effects on communication.
- Include heartbeats, self-test, battery-state validation, and failure propagation in the deployment plan.

Radiation-Shield Geometry and Material Studies

These papers provide the strongest evidence for selecting and comparing passive and active shield designs.

6. Botero-Valencia et al. (2022)

Design and Implementation of 3-D Printed Radiation Shields for Environmental Sensors

DOI: <https://doi.org/10.1016/j.ohx.2022.e00267>

Sensors	Three identical SHT10 temperature/RH sensors; Vantage Pro reference; Particle Argon logger.
Materials	Orange PLA and white ASA; PVC mount; metal spacers; IP69 electronics box.
Geometry	Six truncated cones: five perforated cones and a solid roof, separated by 20 mm spacers.
Experimental setup	Exposed, PLA-shielded, and ASA-shielded sensors compared outdoors, plus 30/90-day weathering tests.

What it shows: Both printed shields reduced radiation error; ASA was favored for outdoor UV resistance and durability.

Brief conclusion: A modular passive cone-stack shield is a defensible low-cost baseline, but material and color were not fully isolated.

Importance to this research: Provides the most direct literature basis for testing printed passive shields and comparing material/weathering behavior.

Things to consider moving forward

- Control color and surface finish when comparing base materials.
- Treat cone count, spacing, and open area as independent geometry variables.

7. Tarara and Hoheisel (2007)

Low-Cost Shielding to Minimize Radiation Errors of Temperature Sensors in the Field

DOI: <https://doi.org/10.21273/HORTSCI.42.6.1372>

Sensors	Same temperature-sensor type used across test and comparison shields.
Materials	Reflective foil insulation with high albedo and low emissivity.
Geometry	Plate, stacked-plate, cone, double-cone, tube, rocket, pagoda, and perforated-tube variants.
Experimental setup	Controlled outdoor comparison under changing solar irradiance and wind.

What it shows: Stacked plates performed best; open bottoms admitted reflected radiation; tube shields needed substantial sidewall perforation.

Brief conclusion: Radiation blocking alone is insufficient: reflected radiation and ventilation area determine passive-shield error.

Importance to this research: Provides the clearest evidence for selecting and parameterizing passive geometry variants.

Things to consider moving forward

- Measure vent/open-area ratio and include reflected ground radiation.
- Avoid open-bottom and weakly perforated tube designs.

8. Holden et al. (2013)

Design and Evaluation of an Inexpensive Radiation Shield for Monitoring Surface Air Temperatures

DOI: <https://doi.org/10.1016/j.agrformet.2013.06.011>

Sensors	Low-cost temperature sensors, including LogTag units.
Materials	White corrugated-plastic sheet.
Geometry	Folded Gill-style series of plates.
Experimental setup	Three-site outdoor comparison against commercial passive shields, RAWS, and aspirated ASOS reference.

What it shows: A roughly USD 3 shield performed similarly to a commercial passive shield, while low-wind full-sun bias remained.

Brief conclusion: Simple, inexpensive geometry can be effective enough for distributed sensing, but passive limitations remain.

Importance to this research: Creates a useful low-cost manufacturability benchmark for the lab's enclosure comparisons.

Things to consider moving forward

- Compare build repeatability, weathering, and cost alongside accuracy.
- Use an aspirated or higher-grade reference to quantify remaining passive bias.

9. Liu et al. (2023)

Design and Experiments of a Naturally-Ventilated Radiation Shield for Ground Temperature Measurement

DOI: <https://doi.org/10.3390/atmos14030523>

Sensors	Pt100 platinum resistance thermometer.
Materials	Silver-coated aluminum outer plates; black inner surfaces; white-resin internal parts.
Geometry	Multilayer shield with two square plates, annular layers, diversion pipe, eight vents, and truncated-cone reflector.
Experimental setup	CFD-informed design followed by outdoor experimental validation.

What it shows: Outer reflectivity, inner surface treatment, venting, and lower reflectors can address direct and reflected radiation together.

Brief conclusion: Mixed surface finishes and guided natural ventilation offer performance gains, with added fabrication and coating-maintenance complexity.

Importance to this research: Expands the paper beyond polymer choice to include surface finish and reflected-radiation management.

Things to consider moving forward

- Treat exterior reflectance and interior finish as separate variables.
- Include vent count/area and coating degradation in the decision framework.

10. García Izquierdo et al. (2024)

COAT Project: Intercomparison of Thermometer Radiation Shields in the Arctic

DOI: <https://doi.org/10.3390/atmos15070841>

Sensors	Forty-one calibrated four-wire Pt100 thermometers; ultrasonic anemometers and reference meteorology.
Materials	Ten commercial shield models; detailed proprietary materials vary.
Geometry	Natural and active shields, including multiplate and Stevenson-screen configurations.
Experimental setup	Fourteen-month Arctic intercomparison with pre/post calibration.

What it shows: Shield differences depend strongly on wind and radiation; fan, logger, power, memory, mounting, and storm failures caused data loss.

Brief conclusion: Long-duration comparison must evaluate operational reliability as well as thermometer error.

Importance to this research: Provides a strong precedent for long-term failure logging and passive-versus-active comparison.

Things to consider moving forward

- Log fan operation, storage capacity, power status, and mount condition.
- Use redundant probes where possible to distinguish shield effects from sensor failure.

11. Deford et al. (2025)

Development of a Low-Cost, 3D-Printed, Aspirated Air Temperature Measurement Radiation Shield

DOI: <https://doi.org/10.1175/JTECH-D-24-0006.1>

Sensors	Compact air-temperature sensor; Sunon 5 V fan.
Materials	White PLA; hot glue during testing; fan and exposed circuits.
Geometry	Three-part double-wall 90-degree elbow with insect screen, central supports, and angled fan diffuser.
Experimental setup	Outdoor comparison with passive and commercial aspirated reference shields.

What it shows: Low-cost aspiration can achieve strong accuracy, but the fan consumes 680 mW and creates durability and maintenance risks.

Brief conclusion: Active airflow is a useful accuracy benchmark and a direct test of the accuracy-autonomy tradeoff.

Importance to this research: Provides the clearest candidate active design for comparison with passive lab-box options.

Things to consider moving forward

- Measure fan power, flow, and failure behavior.
- Replace temporary PLA/glue/exposed circuitry with weatherable materials and permanent assembly for long deployment.

12. Jin et al. (2026)

Design and Experimental Validation of a High-Accuracy Naturally Ventilated Radiation Shield

DOI: <https://doi.org/10.3390/atmos17030272>

Sensors	Thin-film Pt100 in an 8 mm copper spherical shell.
Materials	CFD comparison of plastic, wood, Fe-Ni alloy, and aluminum; mirror-finish aluminum considered.
Geometry	Two square shading plates with symmetric bowl-cover flow-guiding shrouds and 30 mm spacing.
Experimental setup	CFD material/geometry optimization followed by outdoor validation.

What it shows: Material thermal properties and guided passive airflow jointly affect radiation error.

Brief conclusion: Advanced passive geometry may reduce error without fan power, but complexity and weathering must be assessed.

Importance to this research: Supports evaluating airflow guidance and upward/reflected radiation rather than only plate count.

Things to consider moving forward

- Balance modeled accuracy against manufacturability, size, mounting, and coating durability.
- Validate CFD-derived designs across multiple weather states.

Calibration, Drift, and Air-Quality Sensor Studies

These papers define calibration, drift, environmental correction, and long-term validation requirements.

13. deSouza et al. (2022)

Calibrating Networks of Low-Cost Air Quality Sensors

DOI: <https://doi.org/10.5194/amt-15-6309-2022>

Sensors	Canary-S boxes using Plantower PMS5003 PM sensing with integrated temperature/RH.
Materials	Commercial outdoor box; material and internal airflow not reported.
Geometry	Commercial networked box; detailed inlet/outlet and internal layout not reported.
Experimental setup	Twenty-four school deployments; seven boxes co-located with five FEM reference sites.

What it shows: Calibration evaluated at one co-location may fail to transfer across time or space.

Brief conclusion: Calibration performance must be tested beyond the location and period used to fit the model.

Importance to this research: Directly supports time-separated and, if possible, site-separated validation of the lab box.

Things to consider moving forward

- Document PM inlet, fan path, sensor placement, and enclosure material.
- Report calibration transfer performance rather than only training/co-location error.

14. Clements et al. (2017)

Low-Cost Air Quality Monitoring Tools: From Research to Practice

DOI: <https://doi.org/10.3390/s17112478>

Sensors	Optical PM, electrochemical gas, metal-oxide gas, and multi-sensor systems discussed generally.
Materials	No single enclosure material reported.
Geometry	No single geometry; air exchange and internal heat are identified as important.
Experimental setup	Workshop summary synthesizing laboratory, co-location, mobile, community, and network experience.

What it shows: Co-location, repeated calibration, drift checks, and environmental measurements are common requirements.

Brief conclusion: Low-cost sensor boxes need a defined quality-assurance workflow rather than a one-time validation.

Importance to this research: Provides broad best-practice justification for calibration and enclosure-condition logging.

Things to consider moving forward

- Measure internal heat and air exchange rather than treating the box as neutral.
- Place environmental-correction sensors close to pollutant sensors while exposing them to ambient air.

15. Giordano et al. (2021)

From Low-Cost Sensors to High-Quality Data

DOI: <https://doi.org/10.1016/j.jaerosci.2021.105833>

Sensors	Low-cost optical particulate-matter sensors as a class.
Materials	No single enclosure material reported.
Geometry	No single enclosure; packaging and airflow identified as influences.
Experimental setup	Review of laboratory evaluation, co-location, and network deployment studies.

What it shows: PM response depends on humidity, aerosol properties, sensor age, site, and packaging/airflow.

Brief conclusion: PM calibration and enclosure design are deployment-specific and cannot be treated independently.

Importance to this research: Defines environmental covariates and limitations that must be considered if the lab box includes PM sensing.

Things to consider moving forward

- Include humidity, inlet airflow, aging, and local aerosol conditions in PM analysis.
- Do not infer a universally transferable calibration.

16. Vajs et al. (2021)

Developing Relative Humidity and Temperature Corrections for Low-Cost Sensors Using Machine Learning

DOI: <https://doi.org/10.3390/s21103338>

Sensors	Alphasense gas cells; Bosch BME280; Plantower PMS7003.
Materials	Commercial AQ10x enclosure; material and airflow not reported.
Geometry	Commercial outdoor air-quality station; internal layout not reported.
Experimental setup	One station co-located with a Serbian public monitoring station; LR, MLR, and ML corrections compared.

What it shows: Temperature and RH correction can improve low-cost gas/PM measurements, but sensor-specific sensitivity remains.

Brief conclusion: Interpretable linear and multiple-linear models provide a defensible baseline before more complex methods.

Importance to this research: Directly supports the proposed calibration sequence for the lab box.

Things to consider moving forward

- Start with linear and multiple-linear calibration and use a time-separated test period.
- Ensure T/RH sensors measure the air reaching the pollutant sensors, not only the electronics-box microclimate.

17. Schneider et al. (2023)

Deployment and Evaluation of a Network of Open Low-Cost Air Quality Sensor Systems

DOI: <https://doi.org/10.3390/atmos14030540>

Sensors	Alphasense NO ₂ , CO, NO, O ₃ ; ELT D-300 CO ₂ ; OPC-N3 PM; T/RH/pressure.
Materials	AirSensEUR boxes; enclosure material not reported.
Geometry	Complete boxes co-located together, then distributed to roadside sites.
Experimental setup	Five-week pre-deployment co-location, roadside deployment, and post-deployment evaluation.

What it shows: High RH affected optical PM; unit-to-unit differences, drift, and failures required pre/post deployment evaluation.

Brief conclusion: Pre- and post-deployment co-location improves detection of drift and field-induced change.

Importance to this research: Supports comparing multiple lab boxes and checking calibration after deployment.

Things to consider moving forward

- Perform pre/post co-location if deployment duration permits.
- Treat humidity, unit variability, and multi-sensor maintenance as primary factors.

18. Diez et al. (2024)

Long-Term Evaluation of Commercial Air Quality Sensors: The QUANT Study

DOI: <https://doi.org/10.5194/amt-17-3809-2024>

Sensors	Forty-three commercial devices containing 119 gas and 118 PM sensors.
Materials	Commercial protective cases; materials and internal layouts generally proprietary.
Geometry	Multiple commercial black-box systems.
Experimental setup	Three-year evaluation across UK urban environments, including relocation and uncertainty analysis.

What it shows: Performance varied substantially between systems; relocation and manufacturer correction changes affected results.

Brief conclusion: Long-term validation requires documentation of hardware, firmware, processing, location, and uncertainty.

Importance to this research: Supports treating the lab box as a transparent system and evaluating more than RMSE alone.

Things to consider moving forward

- Document every hardware, firmware, and calibration change.
- Evaluate inter-device precision, uncertainty, relocation, and long-term stability.

19. Winter et al. (2025)

Sustained Performance of Low-Cost Air Quality Sensors in Long-Term Deployments

DOI: <https://doi.org/10.1021/acssensors.5c00566>

Sensors	Alphasense NO2-B43F, Ox-B431, NO-B4, and CO-B4 electrochemical sensors.
Materials	BEACO2N node enclosure not described in detail.
Geometry	Dense-network node packaging; geometry not the focus.
Experimental setup	Three-year San Francisco Bay Area network evaluation.

What it shows: Calibrated electrochemical performance can remain stable for years, while mechanical/electrical failures may occur first.

Brief conclusion: Recalibration interval and physical-maintenance interval are separate engineering quantities.

Importance to this research: Strengthens the paper's focus on identifying the true limiting factor rather than assuming sensor drift dominates.

Things to consider moving forward

- Track recalibration need separately from physical interventions.
- Record board, connector, power, and enclosure failures alongside sensing-element behavior.

Recommended Experimental Framing

The literature supports an engineering comparison centered on the lab's existing box. The primary result should be identification of the limiting factor: sensor calibration, physical integration, power, firmware/data handling, or maintenance.

Recommended Initial Designs

Design	Purpose
Current lab box	Unmodified baseline that establishes present accuracy, autonomy, and failure modes.
Strong passive option	White reflective stacked-plate or truncated-cone shield with electronics isolated in an IP-rated compartment.
Active benchmark	Aspirated shield used only if the power budget permits; records fan power and fan-operating status.

Measurements Needed

- Reference-matched sensor readings and timestamps.
- Solar exposure or radiation proxy, wind, rain, ambient temperature, and RH.
- Battery voltage/current, fan state if used, resets, storage and communication health.
- Expected and received sample counts by sensor channel.
- Maintenance, calibration, physical inspection, and failure-event log.
- Material, color/finish, geometry dimensions, vent/open-area ratio, and sensor placement.

Questions to Resolve Before Deployment

- What variables must the lab box measure, and what reference instruments are available for each?
- What calibrated-error, completeness, and autonomy thresholds define a useful deployment?
- Which current component is most likely to limit performance: sensor, enclosure, power, firmware, or maintenance?
- Can the lab fabricate ASA reliably, or should PETG/coated PETG and purchased IP-rated boxes be the practical candidates?
- Is active aspiration justified by the expected accuracy improvement and available power budget?
- How long must the deployment run to capture representative solar, wind, humidity, rain, and temperature conditions?

Overall Conclusion

Across the 19 sources, the strongest evidence supports a two-zone outdoor sensor-box architecture: protected electronics and power in a sealed, serviceable enclosure, with ambient sensors placed in an external or separately ventilated shield. A light-colored passive stacked-plate or cone shield is the most defensible low-power baseline. Active aspiration is valuable as an accuracy benchmark but must be evaluated against its power and maintenance cost. The paper should therefore report calibration improvement and physical-design performance together, while using observed intervention-free runtime and data completeness to define autonomy.

The final contribution should be a practical decision framework showing which design factors most affect the lab's sensor box and which changes provide measurable improvement without unnecessary complexity.

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